

Prajjawal Rajput

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SUMMARY

Full-stack Software Engineer skilled in TypeScript, React, Next.js, Node.js, and Python. Experienced in production web applications, AI agents, RAG systems, LangChain, LangGraph, and MCP servers.

EDUCATION

JSS Academy of Technical Education

B.Tech in Computer Science and Engineering — CGPA: 8.01

Noida, Uttar Pradesh

Oct 2023 – Present

Modern Era Public School

Class XII, PCM — 91%

Bijnor, Uttar Pradesh

Apr 2021 – May 2022

INTERNSHIP EXPERIENCE

Barco Electronic Systems Pvt. Ltd

Software Engineering Intern — MX Team (ClickShare & XMS Cloud)

Noida, Uttar Pradesh

Jun 2026 – Present

- Built and shipped a **Universal Web QR Code Scanner** in Angular and TypeScript from POC to production behind a feature flag, supporting onboarding for **1.53M+ ClickShare base units** across 500+ global companies.
- Reduced scan latency from **4+ seconds to under 1 second** using native **BarcodeDetector**, **zxing-wasm** fallback for iOS/Firefox, center-crop preprocessing, contrast enhancement, and adaptive zoom.
- Developing MCP servers for ClickShare/XMS Cloud user workflows, exposing product capabilities as structured AI tools for user-facing automation and support tasks.
- Wrote **Mocha unit tests** for Notification Manager, raising coverage from **64% to 99%** across statements, branches, functions, and lines.

PROJECTS

Pigeon | MERN, Socket.io, Zustand, Cloudinary, FastMCP | GitHub Repo | Live Demo

Jan 2026

- Built a full-stack real-time chat application with private 1-to-1 messaging, live presence tracking, and JWT authentication secured via httpOnly cookies.
- Engineered a Socket.io WebSocket layer with per-socket JWT middleware for low-latency message delivery and real-time presence broadcasting.
- Integrated Arcjet rate limiting/bot protection, Cloudinary uploads, and Mailtrap transactional emails.
- Authored a Python FastMCP server exposing 9 REST APIs as LLM tools, enabling authenticated chat actions from Claude Desktop.

AnyQuery AI | Next.js, FastAPI, LangChain, FAISS, Gemini | GitHub Repo | Live Demo

Jul 2026

- Built a full-stack RAG application that answers questions over YouTube transcripts and PDFs using Next.js, FastAPI, LangChain, and Gemini, delivering grounded answers in **under 3 seconds**.
- Implemented document ingestion with recursive chunking (**1000 chars, 200 overlap**), local **all-MiniLM-L6-v2** embeddings, and FAISS top-**k=4** semantic search across **1000+ chunks** per document.
- Developed a responsive AI chat interface with language selection, PDF uploads, processing states, and Markdown-rendered responses.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Angular, Tailwind CSS, Zustand, Redux, TanStack Query

Backend: Node.js, Express.js, Socket.io, REST APIs, WebSockets

AI & Agent Frameworks: LangChain, LangGraph, MCP (Model Context Protocol), FastMCP, RAG, AI Agents

Databases: PostgreSQL, MongoDB, Drizzle ORM, Prisma ORM

Tools & DevOps: Git, GitHub, Sentry, Cloudinary, Clerk, Polar, WebAssembly

ACHIEVEMENTS

- Solved **400+ problems on LeetCode** and **600+ across platforms** spanning Arrays, Trees, Graphs, DP, and Backtracking